



Volatility of Pfizer Stock (2020-2021)

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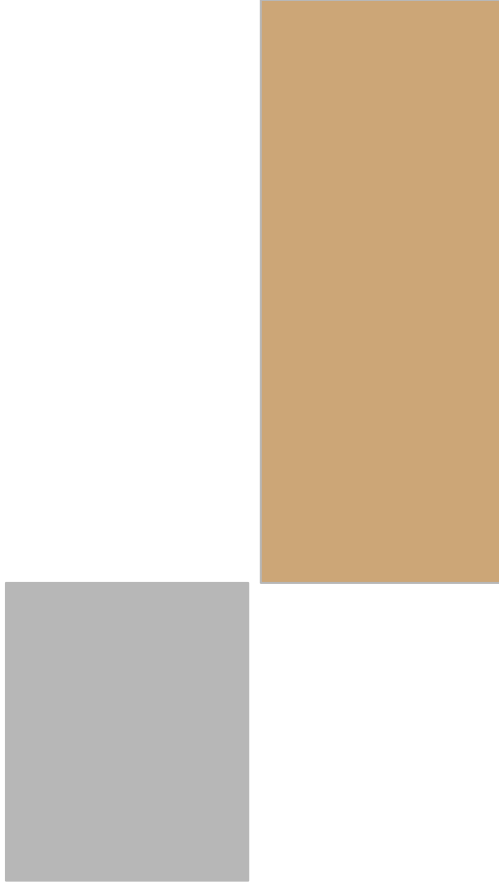
Exploratory Analysis

maps, histograms,
time series analysis,
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01

Problem



Understanding the Problem



Target

Conduct exploratory analysis for Pfizer stocks given the vaccine distribution, adverse reactions, sentiments, and death rates.



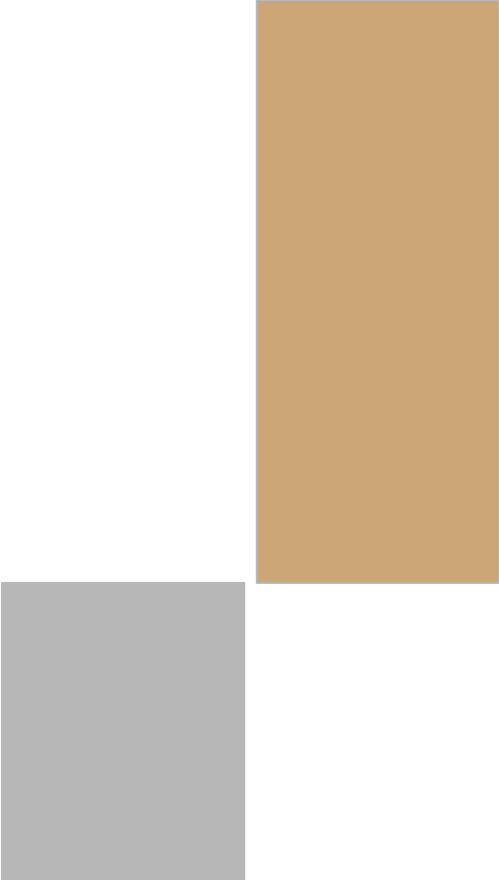
Pfizer

Pfizer is one of the world's largest pharmaceutical company, known for developing vaccines for immunology, oncology, and cardiology.



Consumers

Targets global retailers, hospitals, clinics, government agencies. Nearly, **4.6 billion** vaccines have been shipped to **181 countries**.

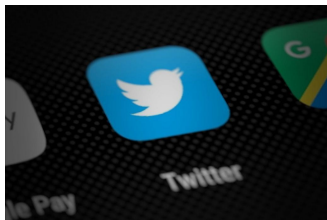


02

Data



Data Overview



Sentiment Data

- Date, Text, Hashtags, Retweets, etc.



Adverse Reaction Data

- State, Symptoms, RecvDate, DateDied, ER_Visit, etc, Vaccination.

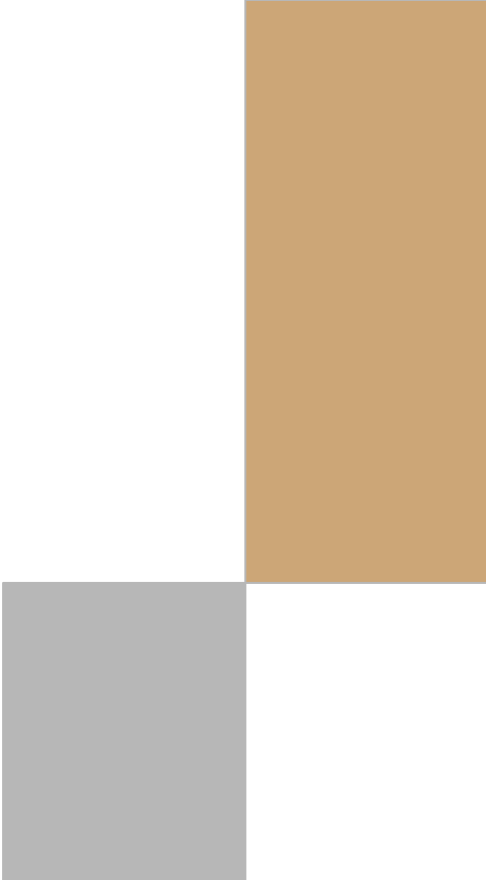


Distribution + Death Data

- Jurisdiction, Allocation Date, Deaths by various cause, etc.



(2020 - 2021)



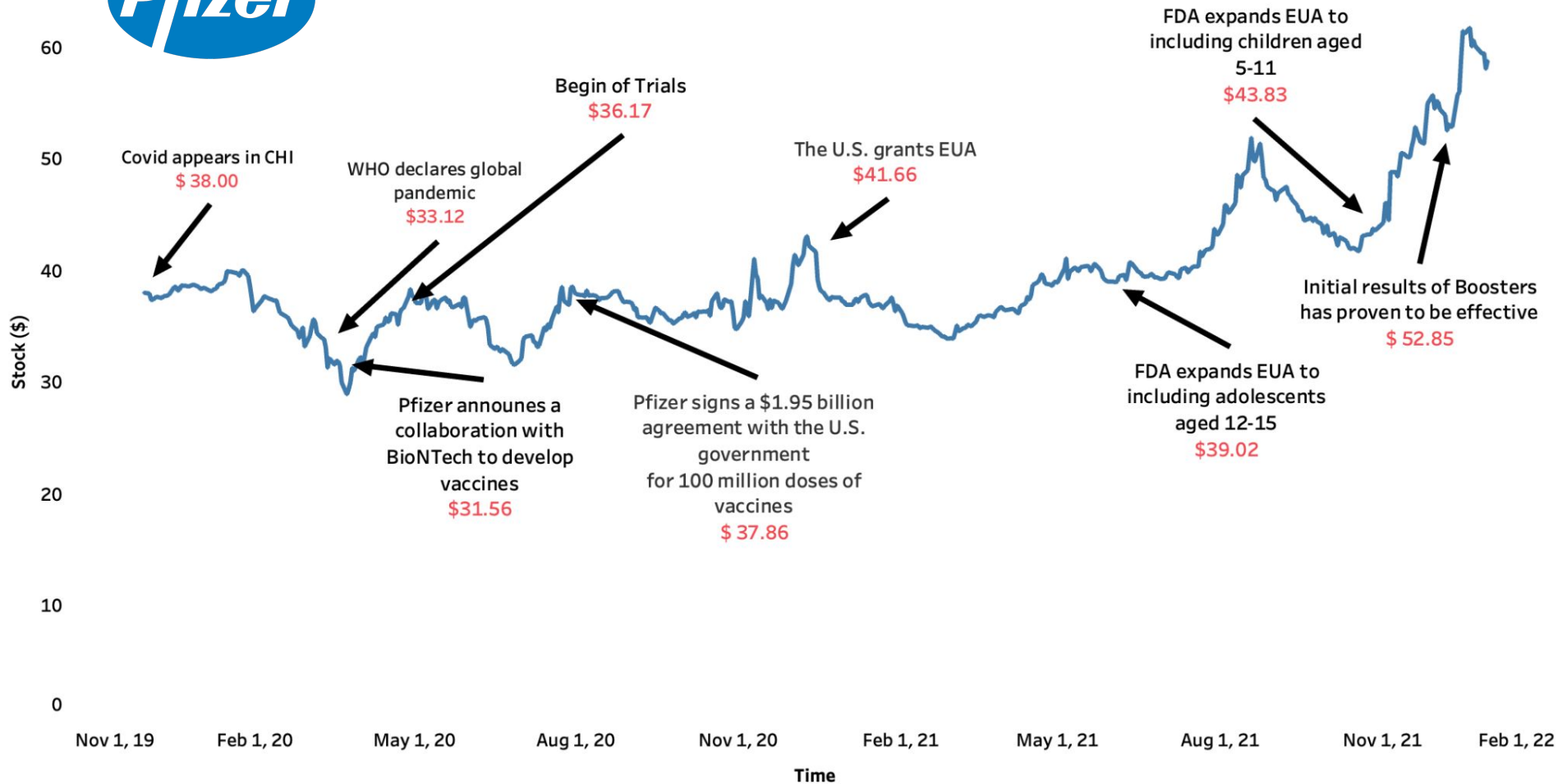
03

Exploratory Analysis

Time Series & Heatmaps, Distribution Plots

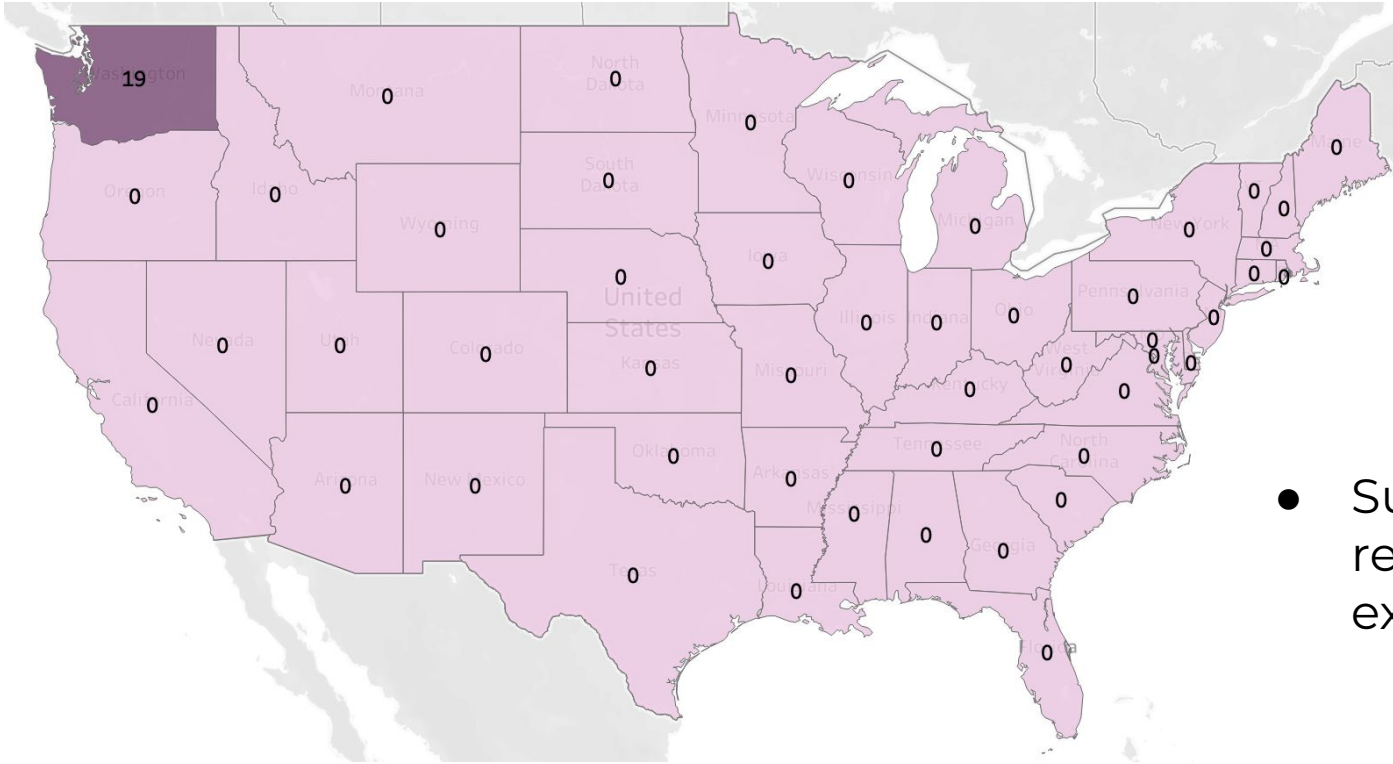


Covid Timeline on Pfizer Stocks





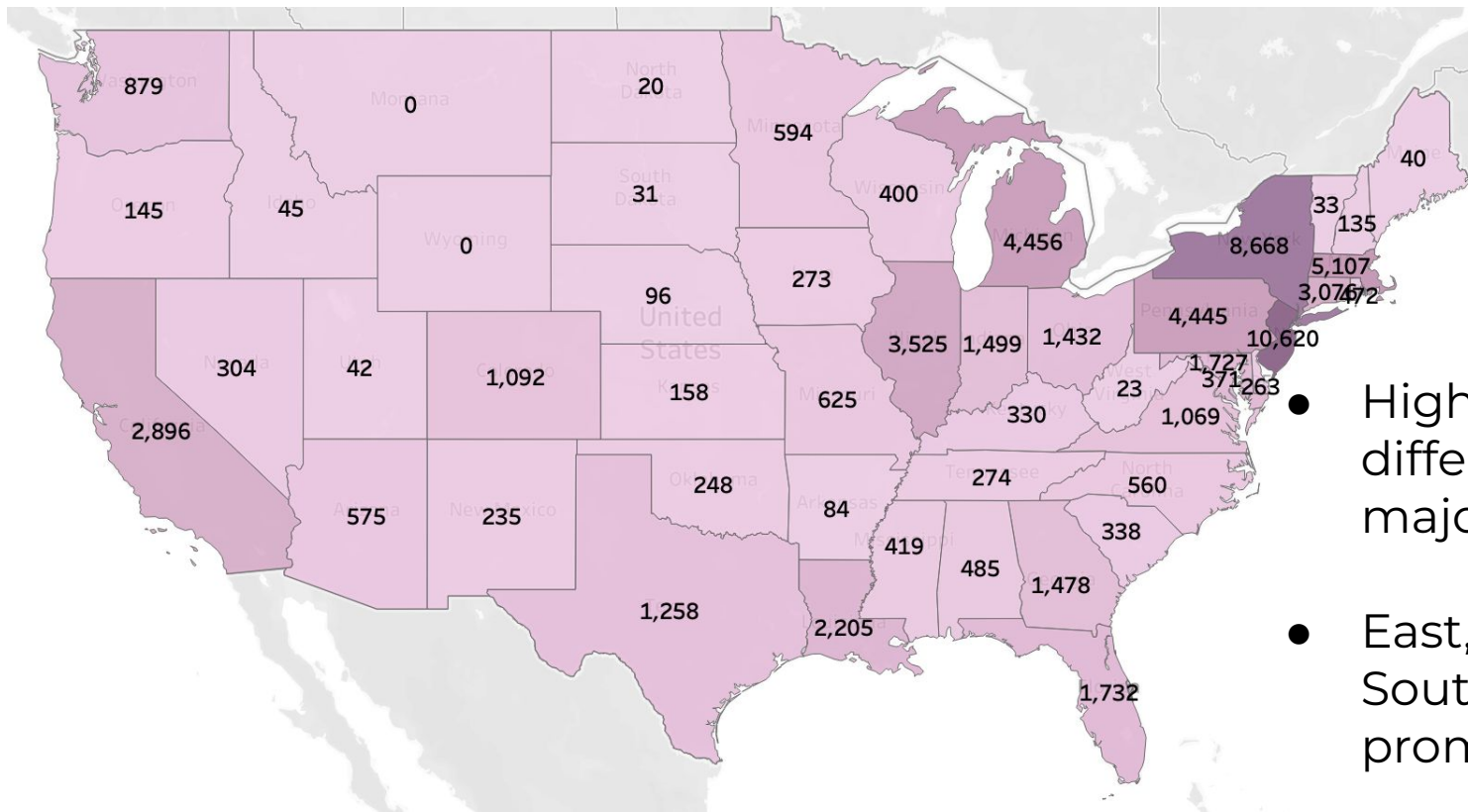
Number of Deaths on 3/11 (WHO declares pandemic)



- Surprisingly no reported deaths except WA



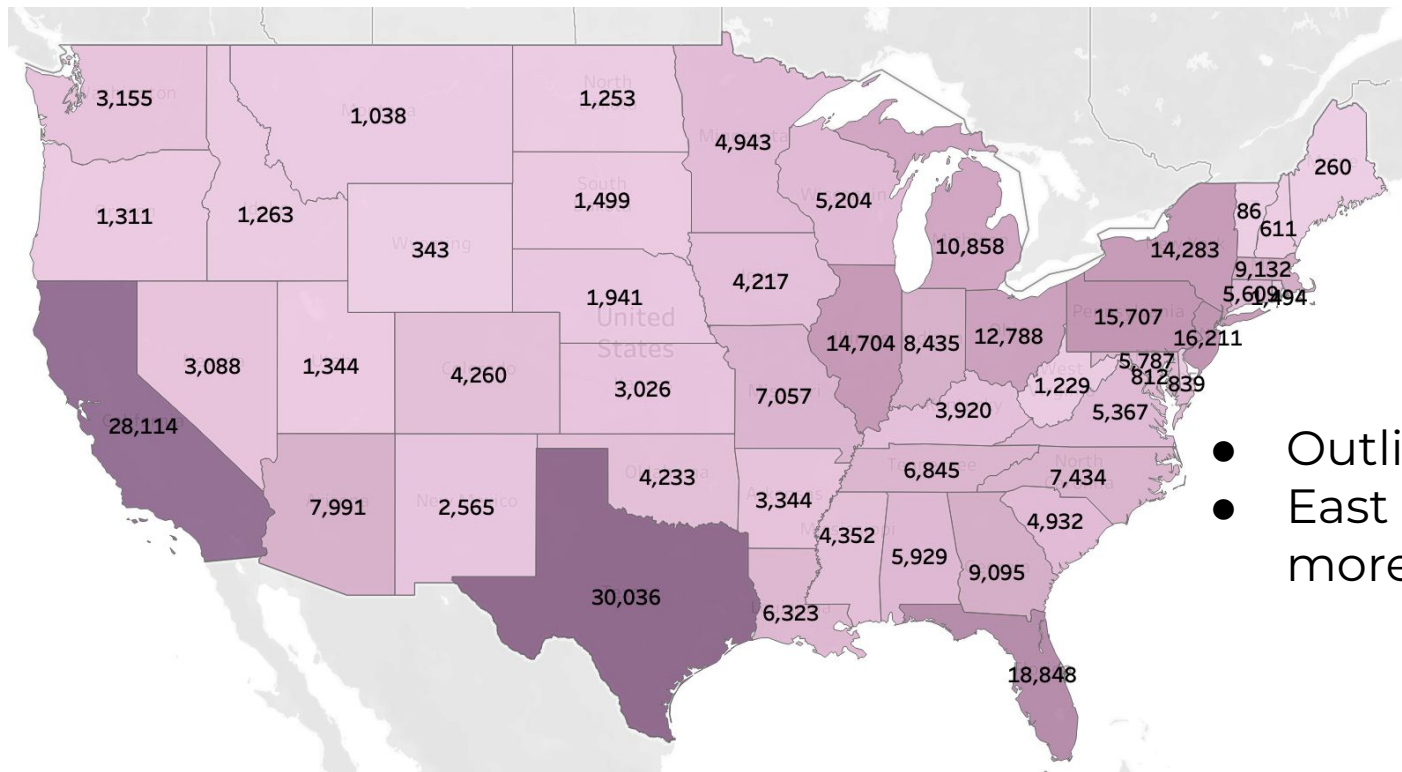
Two Months Later...



- Higher % running difference in majority states.
- East, West, and South appears prominent



By the End of 2020...



- Outliers: CA & TX
- East regions are more concentrated

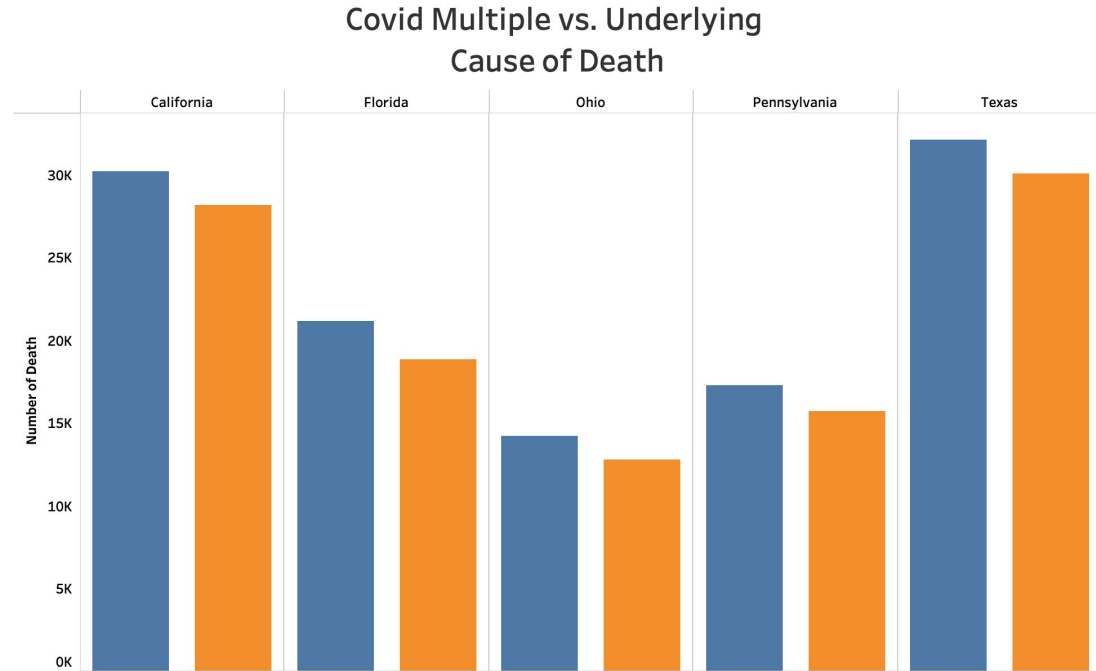


Top 5 States with Most Deaths in 2020

Line: Multiple (not root cause)

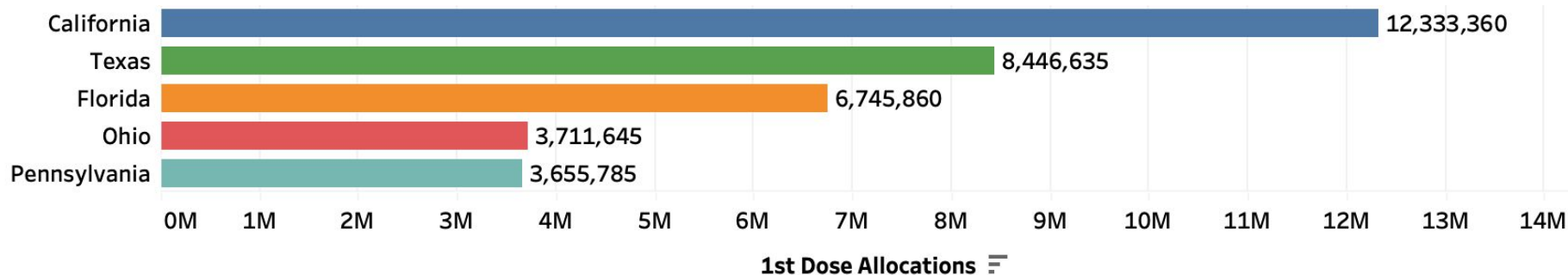
Bar: Underlying (root cause)

- Covid worsened existing health conditions more than being the root cause of deaths.





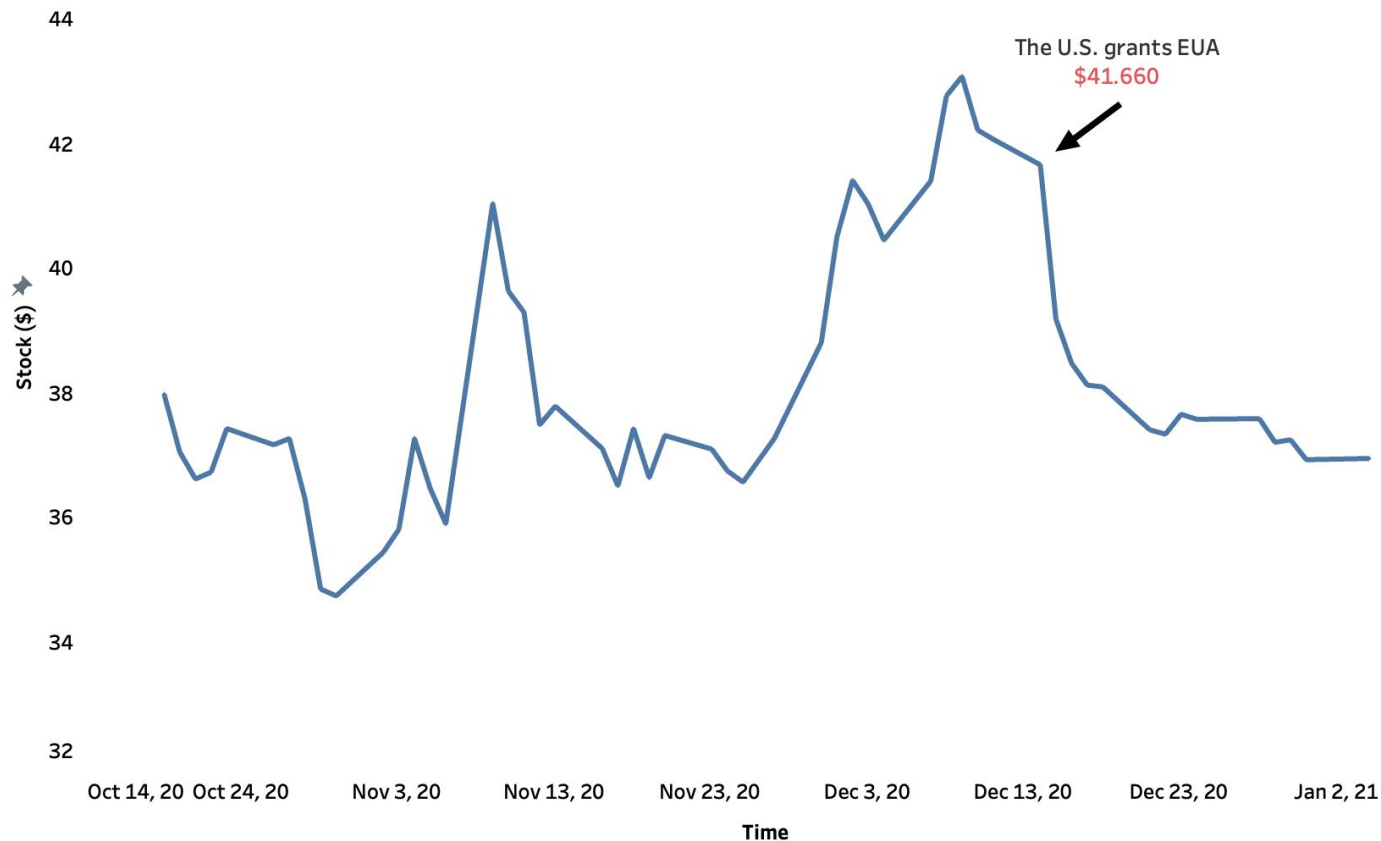
Top 5 States with Most Doses Distributed by Pfizer from late 2020

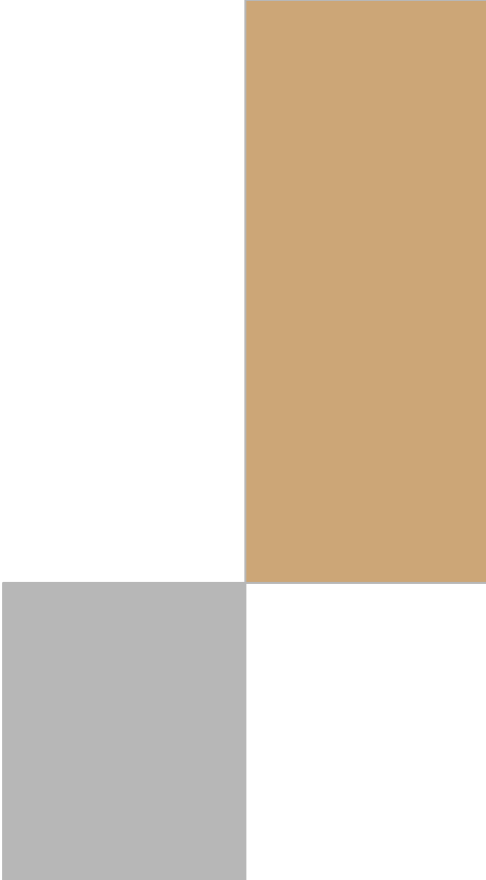


- The Pfizer vaccines were released amidst pandemic (12/11/2020) and their allocation highly corresponded with the number of deaths by states.
- Their vaccines release positively affected their stocks.



Covid Timeline on Pfizer Stocks





03

Exploratory Analysis

Sentiment Analysis



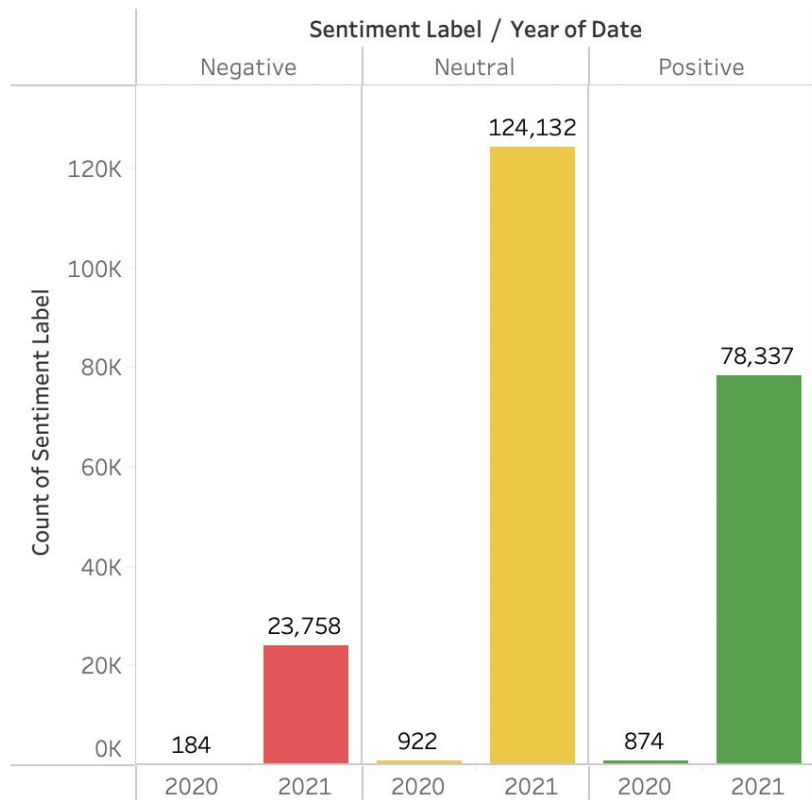


Sentiment Data

Tweet ID	User Name	Tweet Text	Sentiment Score	Sentiment Label
1338548332218630144	Evan Webeck	There is some good news: The first doses of the Pfizer vaccine are being distributed today.	0.316667	Positive
1338322128991641605	Angela Browne	Good news! #PfizerBioNTech vaccines arrived.	0.375000	Positive
1340539111971516416	Rachel Roh	Same folks said daikon paste could treat a cytokine storm...	0.000000	Neutral
1337781178665816064	Braderz73   #GTT	@Pfizer will rake in billions from its expensive vaccine while poor nations struggle to access it.	-0.500000	Negative



Sentiment Analysis



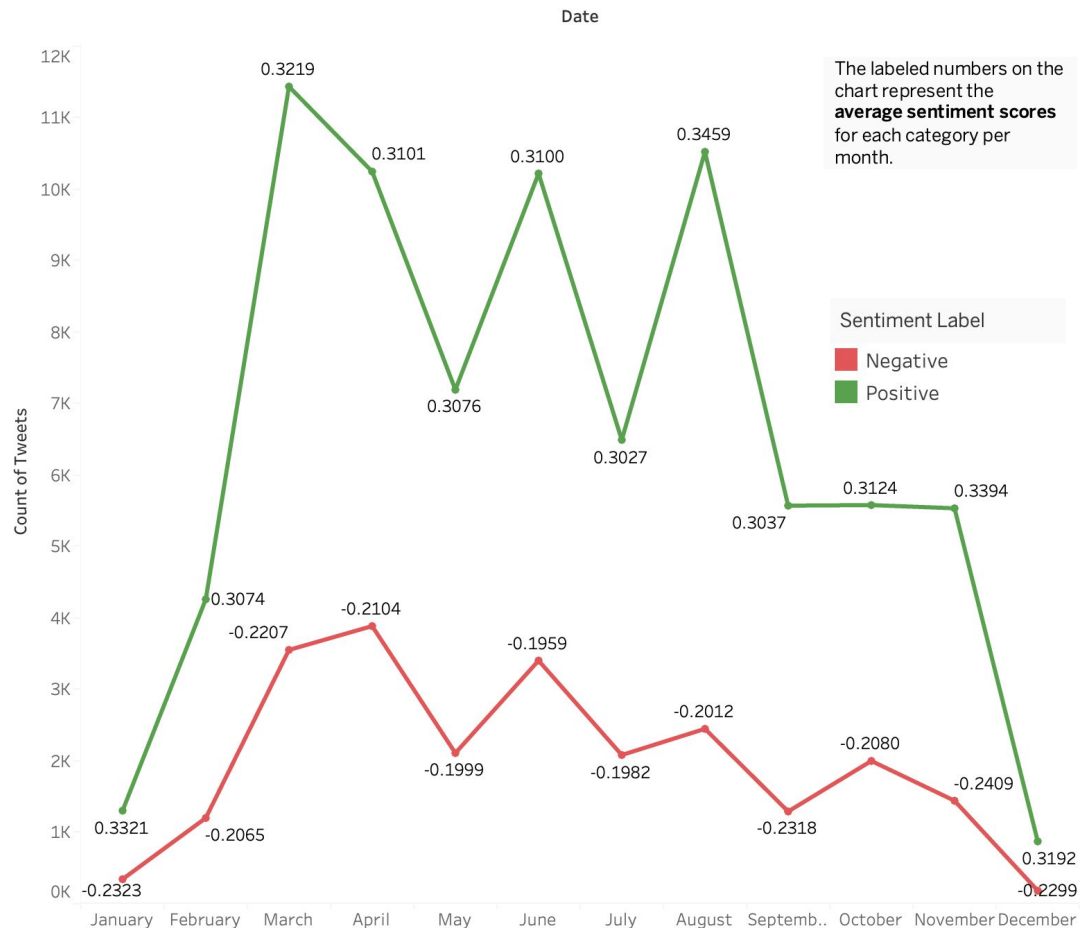
Sentiment Polarity:

- The polarity score ranges from -1 (negative sentiment) to +1 (positive sentiment).
- A score of 0 indicates neutral sentiment.

Sentiment Label:

- Tweets are categorized based on their polarity score:
 - Positive: Polarity > 0
 - Negative: Polarity < 0
 - Neutral: Polarity = 0

Sentiment Timeline with Average Sentiments



- Wide sentiment shifts - the chart shows sentiment spiking from highly positive to deeply negative within just a few months
- Sentiment peaks and valleys - there are clear high and low points in both positive and negative sentiment over the course of the year
- Sentiment extremes at start and end - the most extreme positive and negative sentiment scores occur in the early and late parts of the year



Sentiment Trends:

- **Neutral** dominates, peaking in May, declining after August.
- **Positive** spikes early (Feb-May) but drops post-August.
- **Negative** remains consistently low throughout.

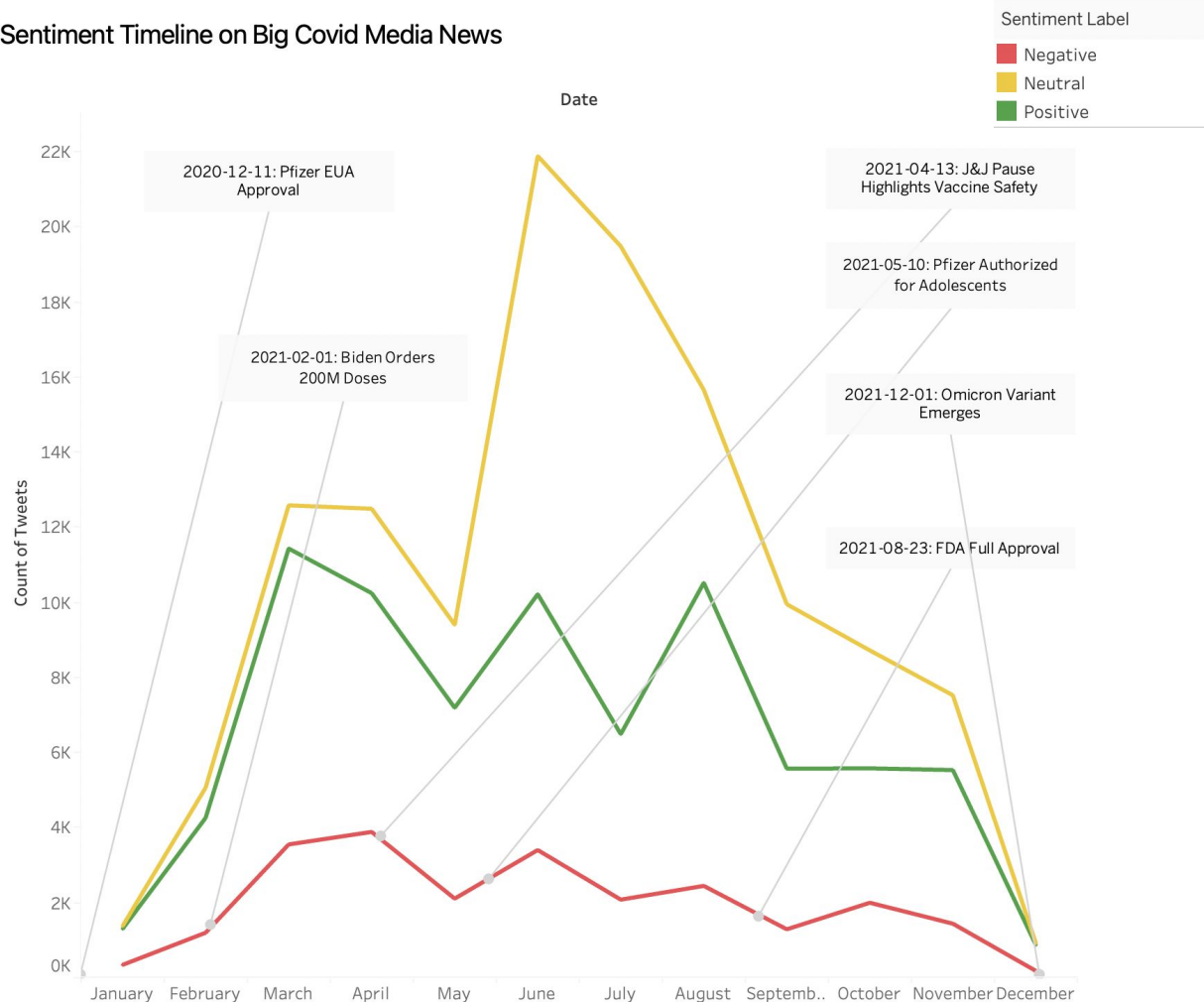
Tweet Volume:

- Peaks align with major events, especially May and December.

Trend Observations:

- Sentiment declines after August, with neutral highest, positive tapering after May, and negative consistently low.

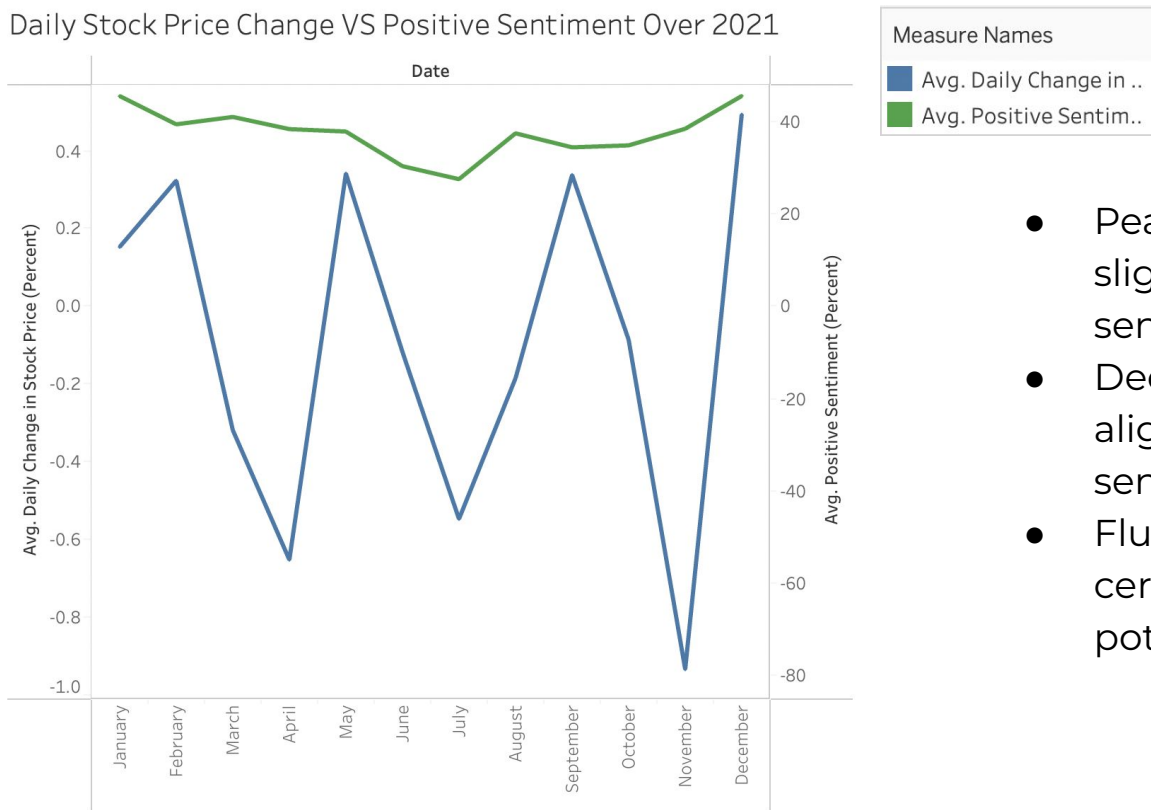
Sentiment Timeline on Big Covid Media News





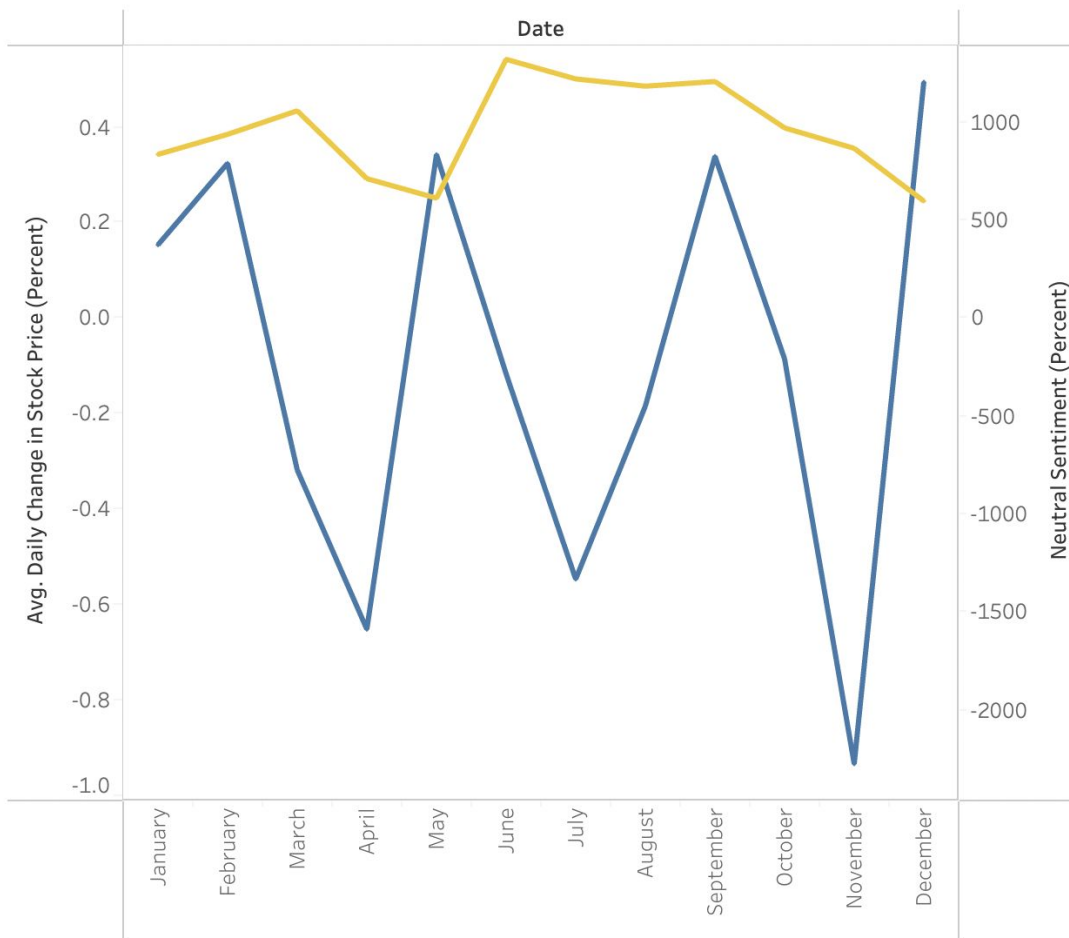
Covid sentiment vs. Stock

Daily Stock Price Change VS Positive Sentiment Over 2021



- Peaks in stock prices match with slight increases in positive sentiment.
- Declines in stock prices tend to align with steady or lower positive sentiment.
- Fluctuations mirror stock trends at certain points, suggesting a potential short-term correlation.

Daily Stock Price Change VS Neutral Sentiment Over 2021

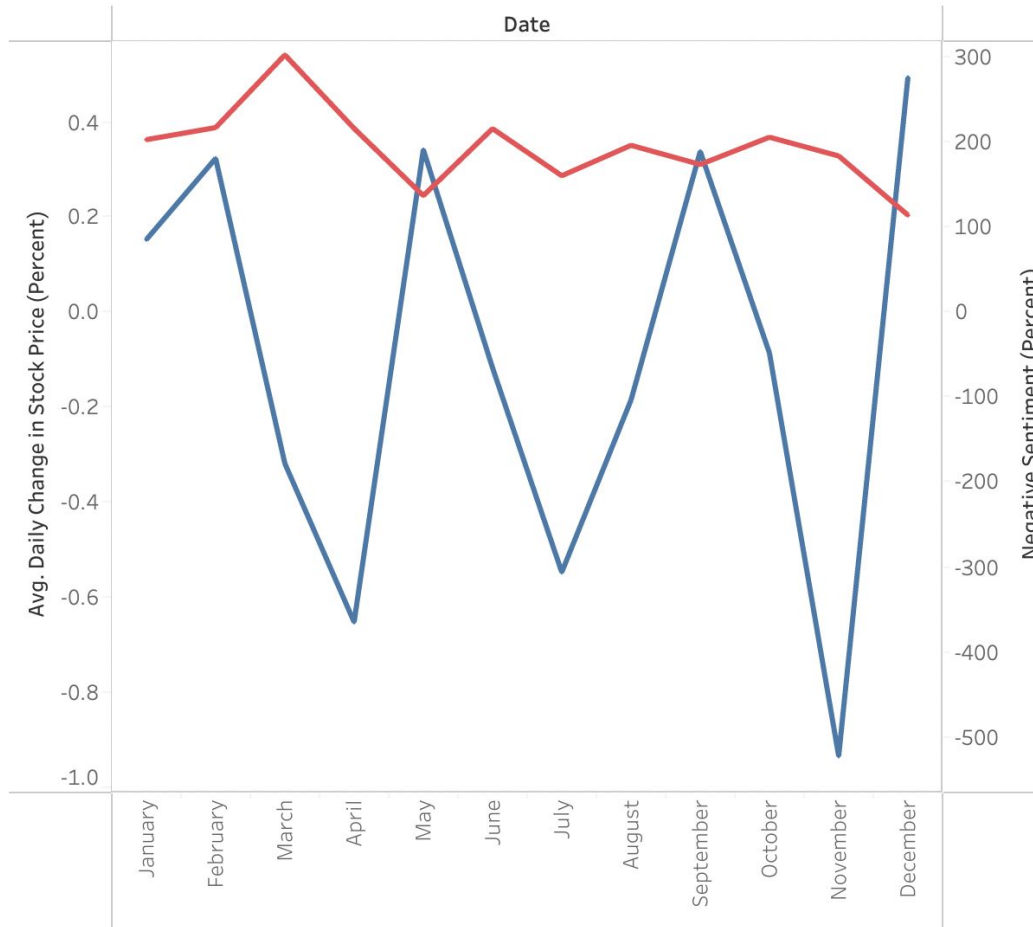


Measure Names

- Avg. Daily Change in ..
- Neutral Sentiment (..

- Shows an inverse relationship between daily stock price changes and neutral sentiment
- Stock price changes and neutral sentiment move in opposite directions, like in March, June, and October
- Both show significant volatility

Daily Stock Price Change VS Negative Sentiment Over 2021

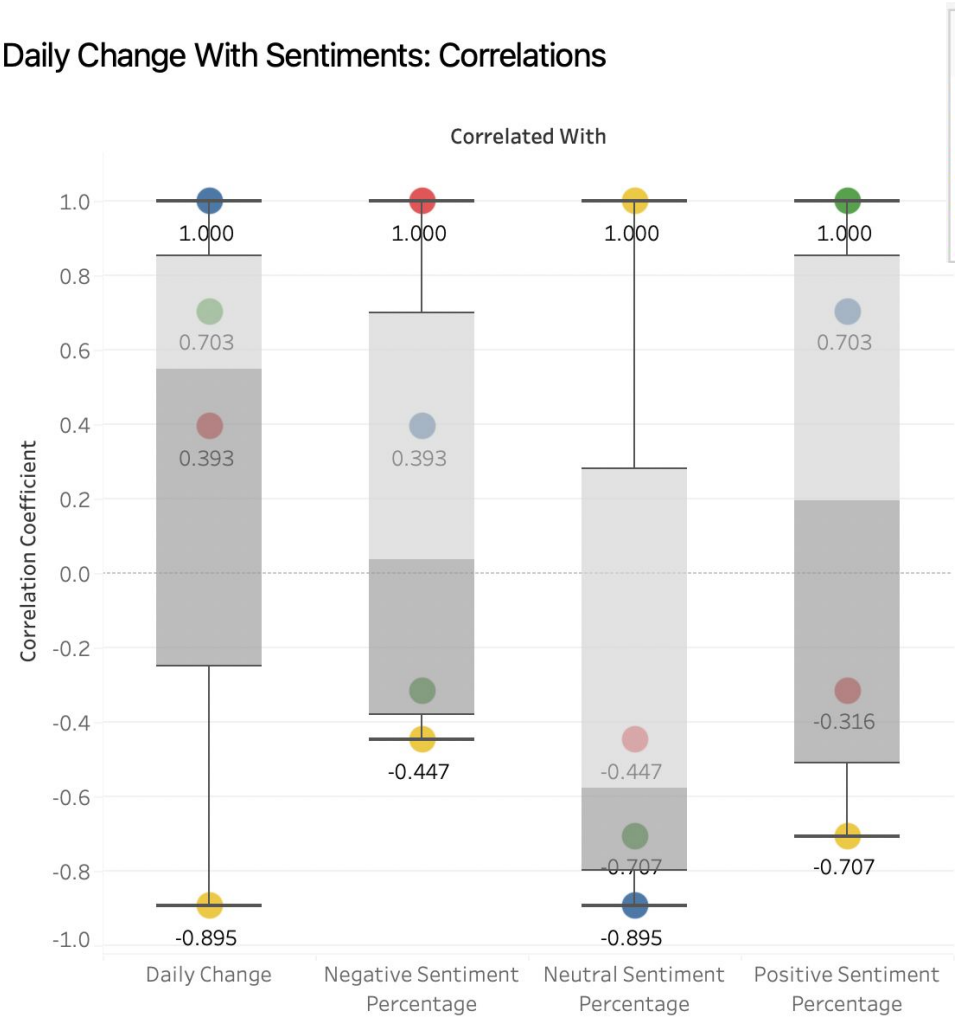


Measure Names

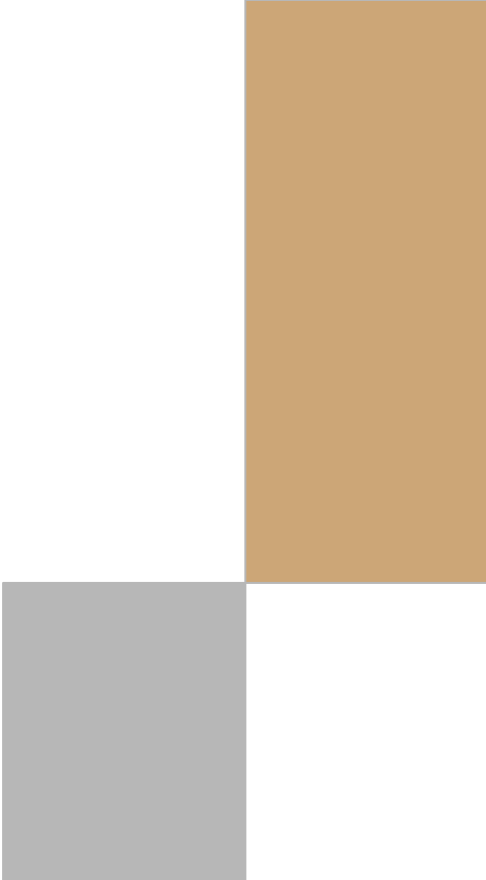
- Avg. Daily Change in ..
- Negative Sentiment ..

- Inverse relationship between stock price changes and negative sentiment.
- Negative sentiment spiked sharply in certain months, even as stock prices were rising.
- High volatility in both across the year

Daily Change With Sentiments: Correlations



- Positive Sentiment:** Strong positive correlation (0.703), indicating stock prices tend to rise with increased positive sentiments.
- Neutral Sentiment:** Strong neutral correlation (-0.895), showing stock prices likely drop significantly on days dominated by neutral sentiments.
- Negative Sentiment:** Weaker negative correlation (-0.447), suggesting stock prices may slightly decline with more negative sentiments.



03

Exploratory Analysis

Adverse Reaction Analysis



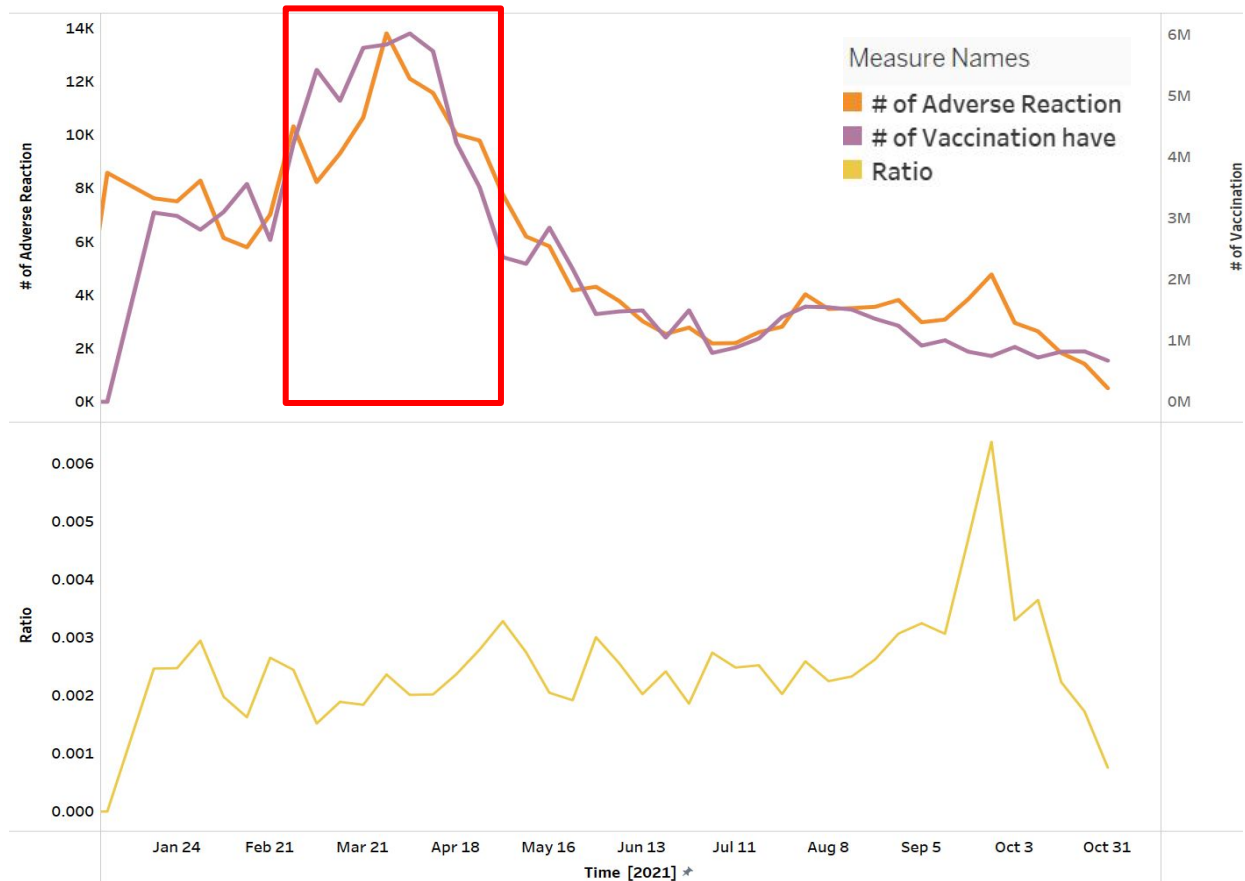


- The trend of adverse reaction numbers is similar to the number of vaccine injections, they have positive relationship.

- The majority of vaccinations occurred between March and April 2021, implying high vaccine demand during that period

- The ratio remained stable for most of the time, suggesting consistent vaccine quality.

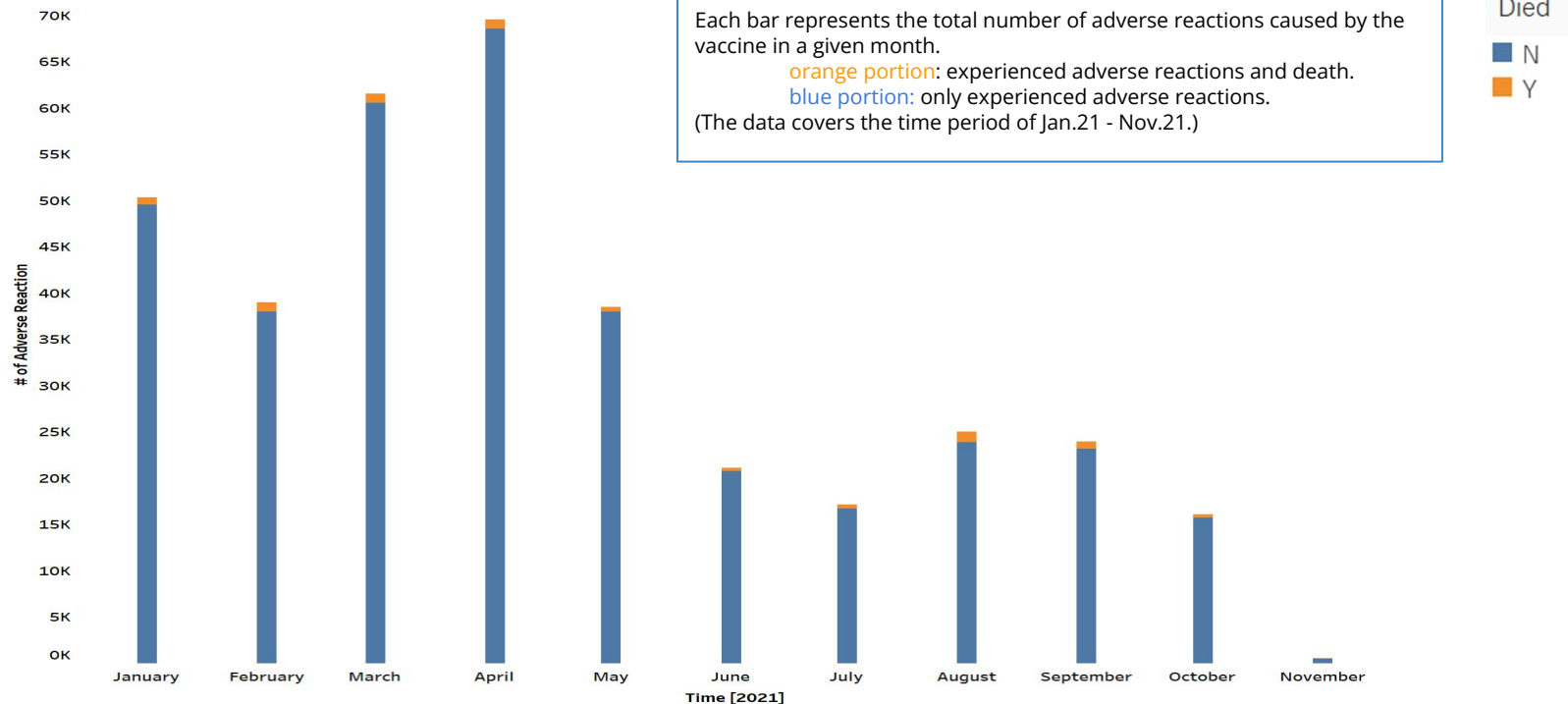
Adverse Reaction & Vaccination



The data covers the time period of Jan.21 - Nov.21.



Adverse Reaction & Death rate



The mortality rate among the population experiencing adverse reactions is close to 2%(7265/362784),
The probability of the vaccine itself causing adverse reactions is very low (close to 0.036%, 362784/100m).



What caused this volatility?

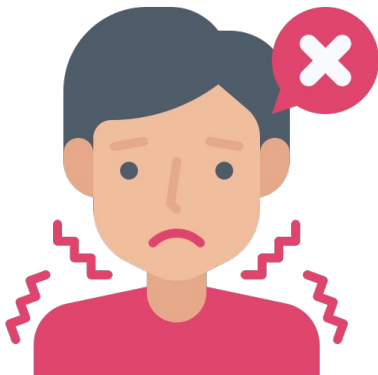
People were suffering:

1. Panic Phase (Dec 2020 – Jan 2021)

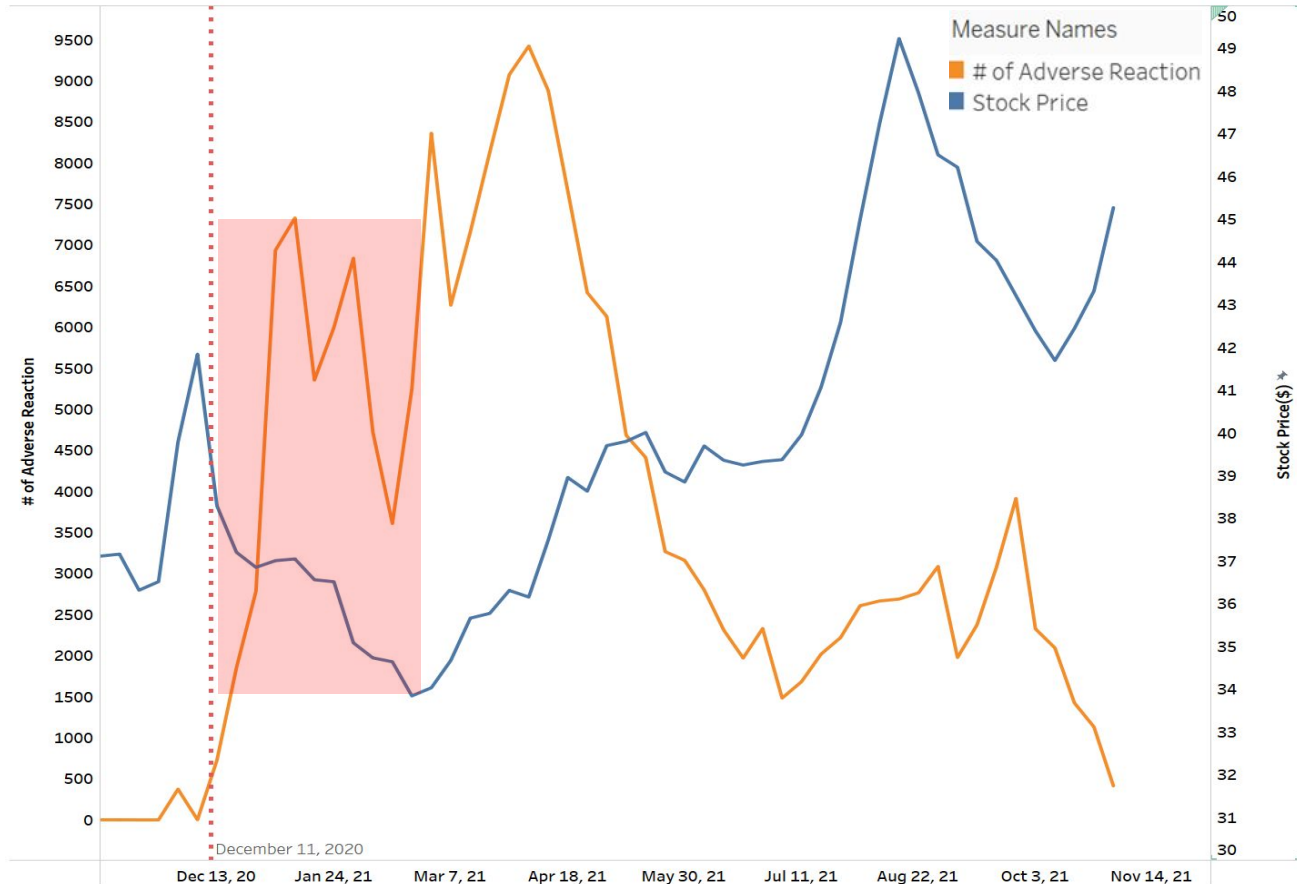
Increase in Adverse Reactions → Stock Price Drops

2. Negative Sentiment (Jan 2021 – Mar 2021)

Decrease in Adverse Reactions → Stock Price Drops Further



Adverse Reaction vs. Stock Price



The data covers the time period of Jan.21 - Nov.21.



What caused this volatility?

People shifted to optimism:

**1. Increased Vaccine Demand
(Mar 2021 – Apr 2021)**

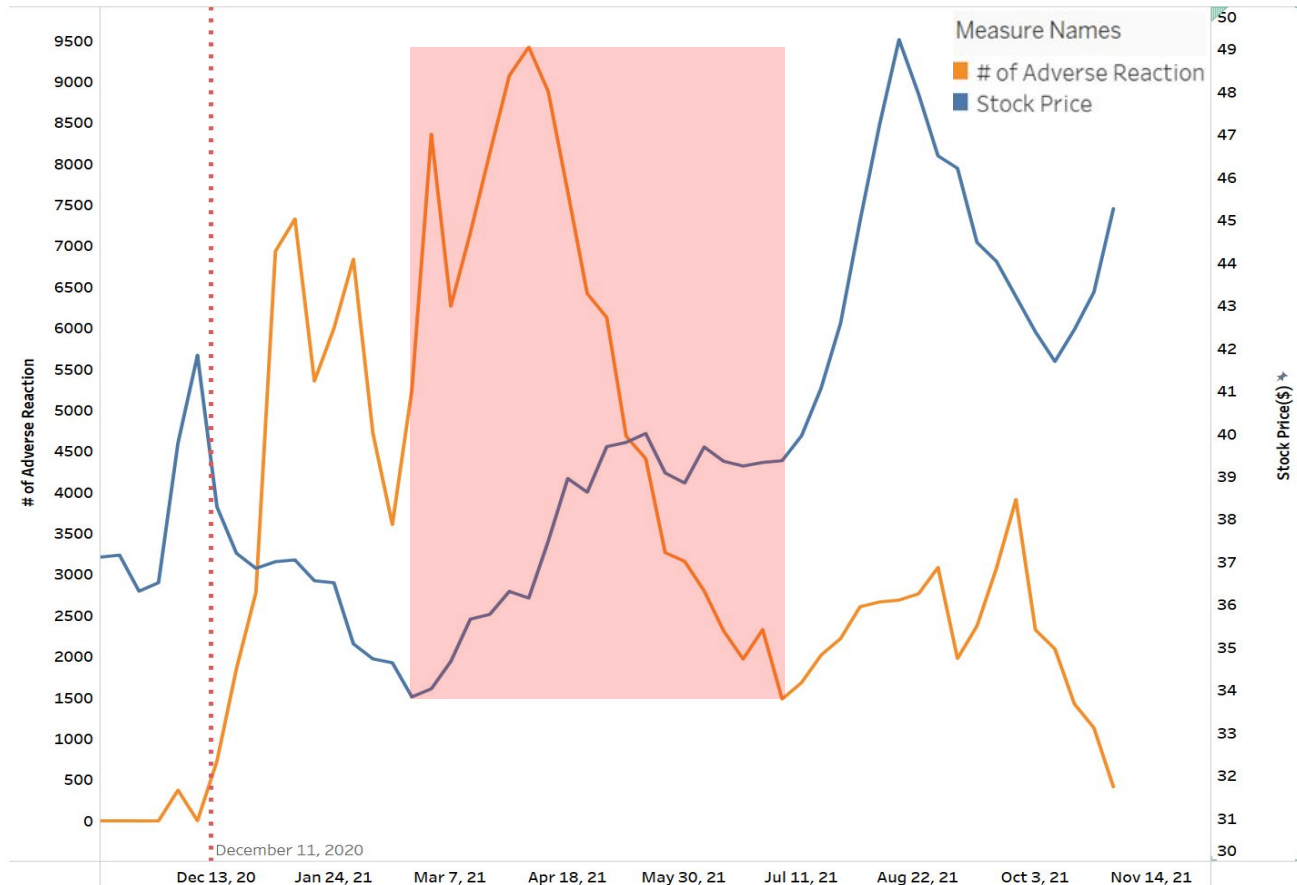
Increase in Adverse Reactions →
Stock Price Rises

**2. Vaccine Safety Validation (Apr
2021 – Jul 2021)**

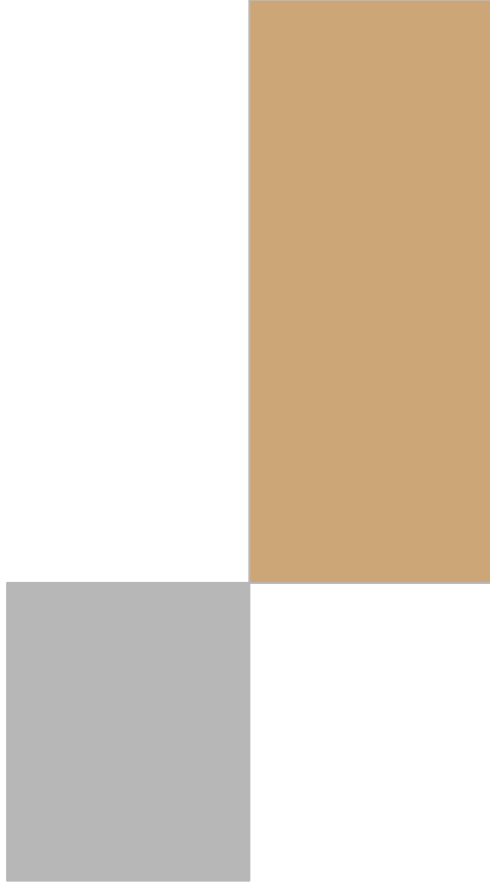
Decrease in Adverse Reactions →
Stock Price Rises Further



Adverse Reaction vs. Stock Price



The data covers the time period of Jan.21 - Nov.21.



04

Conclusion



Impacts & Improvement

For Pfizer:

- Strengthened public trust and global brand reputation.
- Informed decision-making in future health crises.

For Society:

- Saved millions of lives in the world.
- Lost many lives because of vaccines and Covid.
- Findings from this study can inform health agency and policy maker about the impact of public perception on health behaviors and market stability

What we can do next:

1. Explore how the U.S. politics played a role in vaccine distributions and usage (law suits, elections, global vaccine diplomacy, equity debates).
2. Perform competitor analysis (Johnson & Johnson, Moderna, etc)





Limitations

1. **Stock prices are influenced by many external factors.**
 - a. Economic factors (inflation, interests rates)
 - b. Company news & performance
 - c. Industry performance
 - d. Supply and demand
2. **Media coverage:** Public perception and stock prices may be influenced more by media coverage of adverse reactions than by the adverse events themselves.
3. **Data quality:** Incomplete reporting/
lag in data release





References

Nano Pepe. (2021) Covid 19 Vaccines and Deaths by State, Kaggle, Available at:
https://www.kaggle.com/datasets/nanopepe/covid-19-vaccines-and-deaths-by-state?select=COVID-19_Vaccine_Distribution_Allocations_by_Jurisdiction_-_Pfizer.csv

Patrick Gomes. (2020) COVID-19 biotech companies on stock exchange, Kaggle, Available at:
<https://www.kaggle.com/datasets/patrickgomes/pharmaceutical-companies-on-stock-exchange-in-2020>

Levent Serinol. (2021) COVID-19 Vaccine Adverse Reactions (VAERS) Dataset, Kaggle, Available at:
https://www.kaggle.com/datasets/landfallmotto/covid19-vaccine-adverse-reactions-vaers-dataset?select=vaers_jan_nov_2021.csv

Gabriel Preda. (2021) COVID-19 All Vaccines Tweets, Kaggle, Available at:
<https://www.kaggle.com/datasets/gpreda/all-covid19-vaccines-tweets>

Centers for Disease Control and Prevention. (2021) COVID-19 Vaccinations in the United States, Jurisdiction, Available at:
https://data.cdc.gov/Vaccinations/COVID-19-Vaccinations-in-the-United-States-Jurisdiction/unsk-b7fc/about_data

SandhyaKrishnan02. (2021) United States COVID-19 vaccinations Data, Kaggle, Available at:
<https://www.kaggle.com/datasets/sandhyakrishnan02/united-states-covid19-vaccinations>

Thank you

